The Registration of Functional Data

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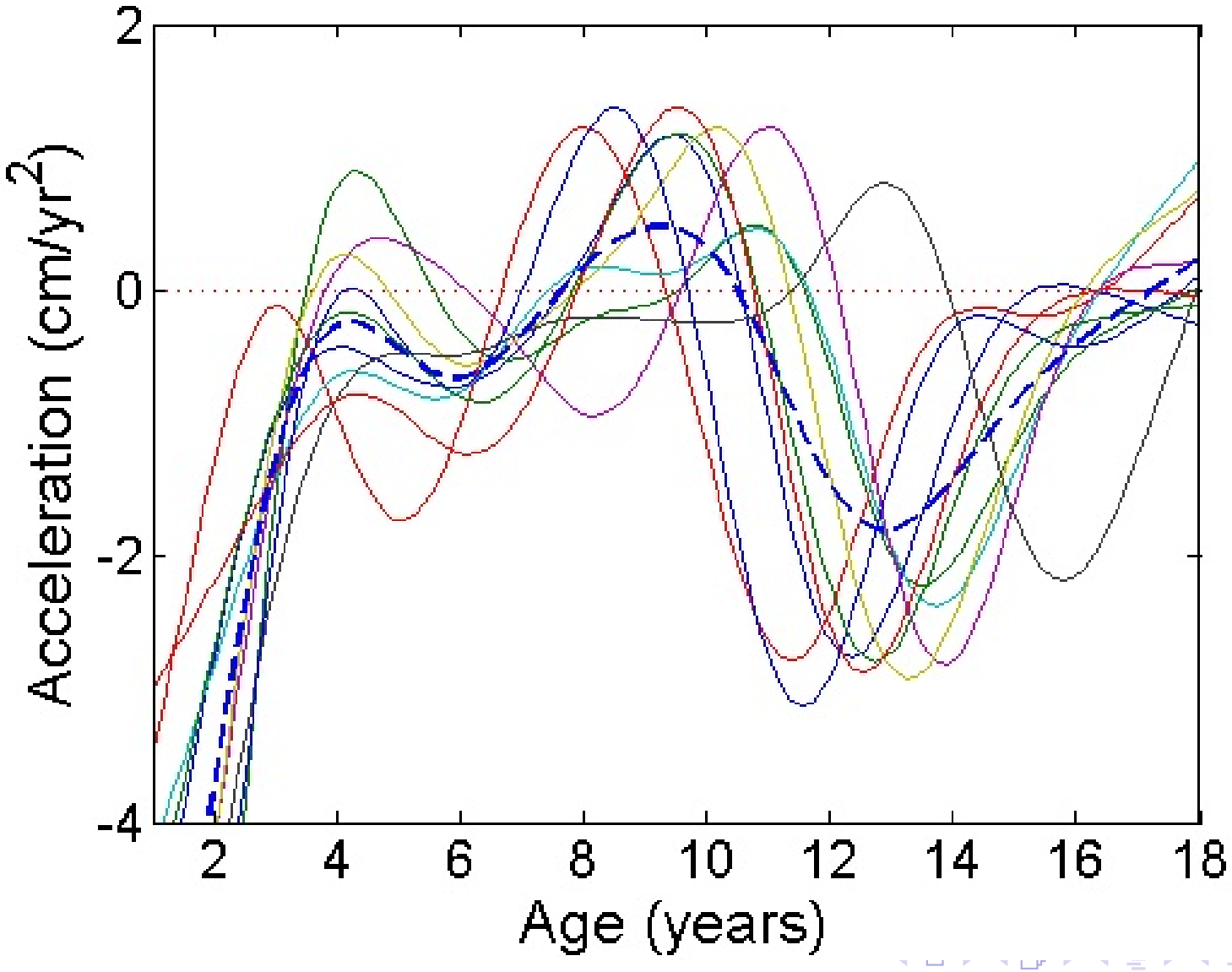
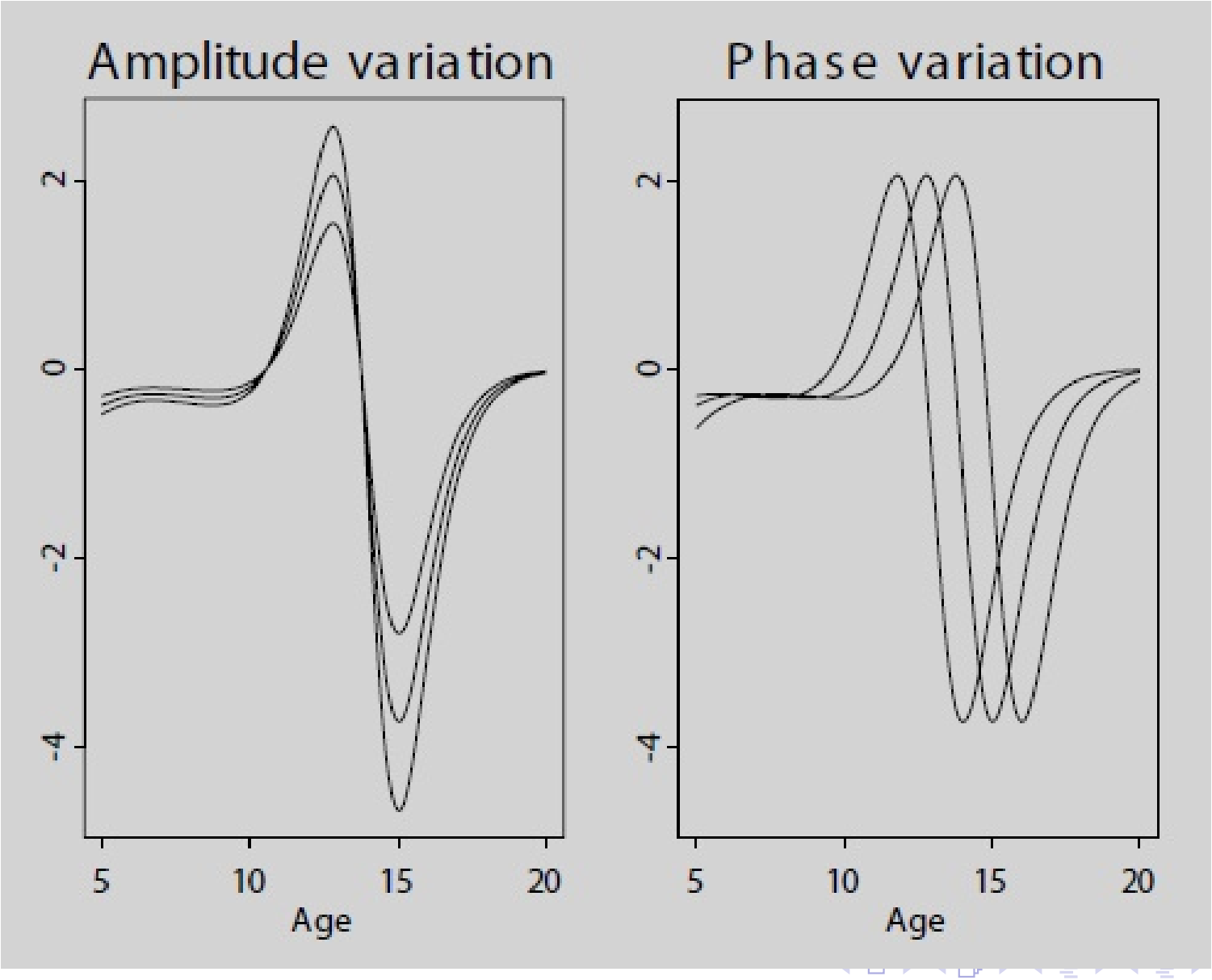
Growth acceleration curves of ten girls



Why curve registration?

- These curves show two types of variation:
  - The usual *amplitude variation*, seen in the intensity of the pubertal growth spurt.
  - Also *phase variation*, visible in the variation in the timing of the pubertal growth spurt.
- When we look at the mean curve, it does not resemble any single curve:
  - the "intensity" of the mean spurt is less than that of any single curve.
  - the "duration" of the mean spurt is greater than that of any single curve.
- We are averaging over units in different states.

Sources of variation



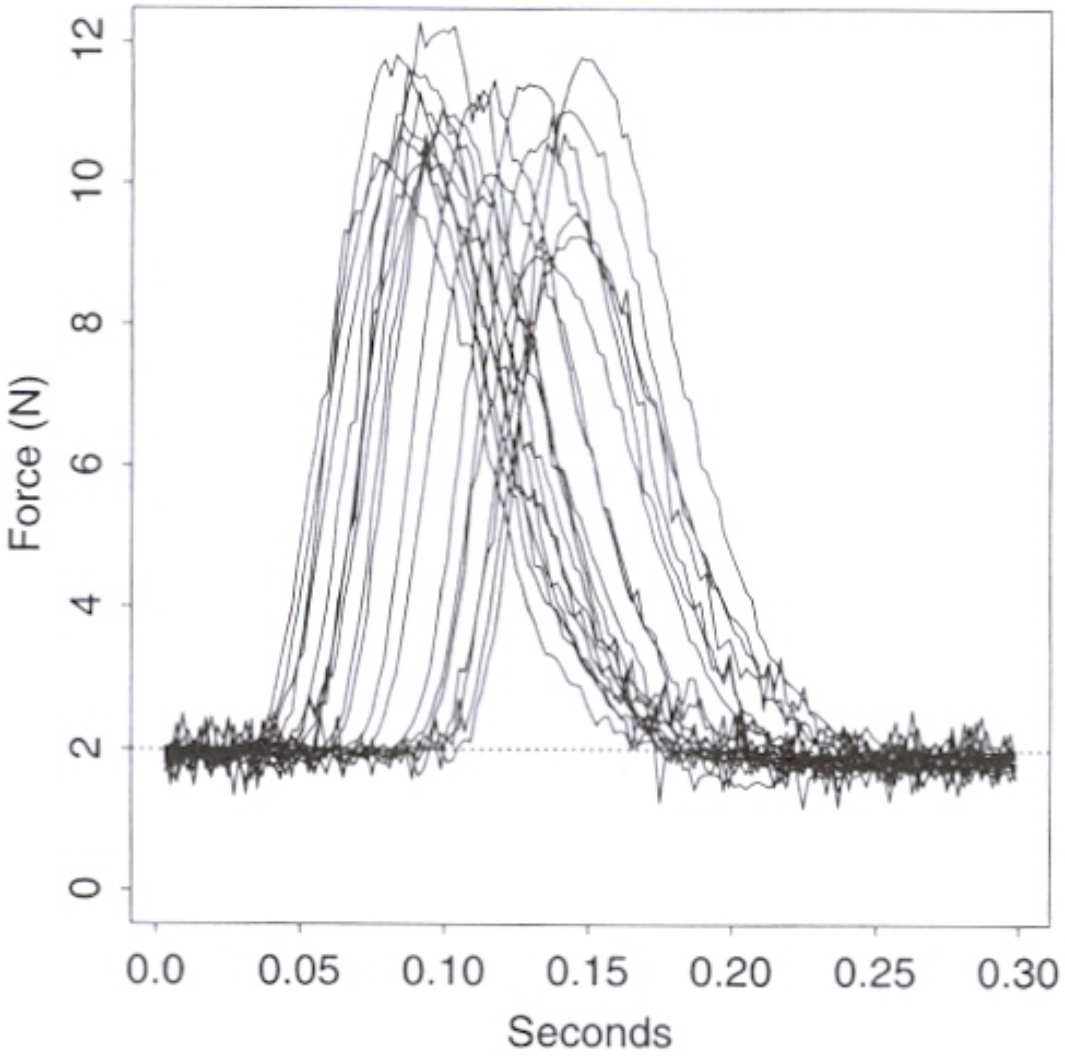
Curve registration

- The need to transform curves by transforming their arguments, which we call *curve registration*, can be motivated as follows.
- The rigid metric of physical time may not be directly relevant to the internal dynamics of many real-life systems.
- Rather, there can be a sort of biological or meteorological time scale that can be nonlinearly related to physical time, and can vary from case to case.
- We contrast *system time* and *clock time*.

The pinch force data

- Reference: Ramsay, Wang and Flanagan (1995). "A functional data analysis of the pinch force of human fingers". *Applied Statistics* **44**: 17-30.
- The data consisted of $N = 20$ records of the force exerted on a meter during a brief pinch by the thumb and forefinger.
- The subject was required to maintain a certain background force on a force meter and then to squeeze the meter aiming at a specified maximum value, returning afterwards to the background level.
- The purpose of the biomechanical experiment was to study the neurophysiology of the thumb-forefinger muscle group.

Pinch force data

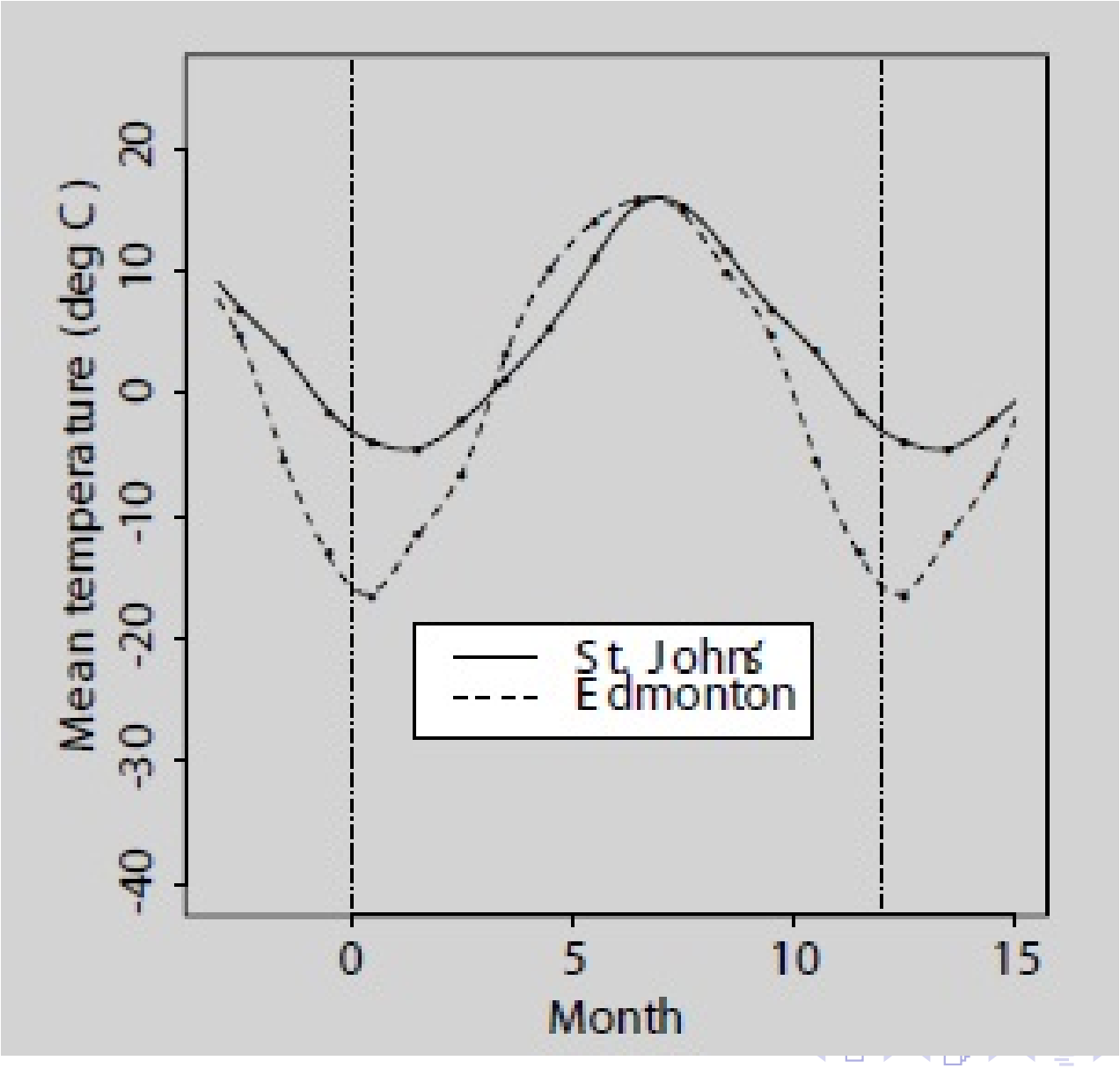


Shifts as nuisance effects

The shift for each individual curve

- measures the gap between the initialization of recording and the beginning of a squeeze,
- is of no real interest,
- and called a *nuisance effect*: a parameter must be accounted for but is of no real interest.

Temperature records at two stations



Shifts as random effects

- Temperature records at different stations can differ noticeably in terms of the phase or timing of key events, such as the lowest mean temperature and the timing of the seasons.
- The shifts to align the curves are of intrinsic interest and should be viewed as a component of variation that needs careful description.
- For example, continental stations such as Edmonton have earlier seasons than marine stations such as St. John's, due to the capacity of oceans to store heat and to release it slowly.
- Such a shift is an important feature of the curve and characterized as a *random effects* problem.

Notation

- Let the interval $\mathcal{T}$ over which the functions are to be registered be $[T_1, T_2]$.
- Assume that each curve x_i is available beyond each end of $\mathcal{T}$.
- We want the values

$$x_i^*(t) = x_i(t + \delta_i),$$

where the *shift parameters* δ_i align the curves. Curves $x_i^*(t)$ are *registered* by shifts δ_i.

Two strategies for alignment

- Strategy one:
  - We can align curves by fixing the location of a feature, such as the summer maximum or winter minimum.
  - This works provided the location of this feature is easy to determine in each curve.
- Strategy two:
  - We can also align curves by using the entire curve.
  - This is always possible, but needs an explicit criterion for alignment.

The least square criterion for shift alignment

- First we estimate a mean function $\hat{\mu}(t)$ for t in $\mathcal{T}$.
- If the individual functional observations x_i are smooth, we can estimate $\hat{\mu}$ by the sample average $\bar{x}$.
- Then we can minimize the *least squares criterion* with respect to δ_i:

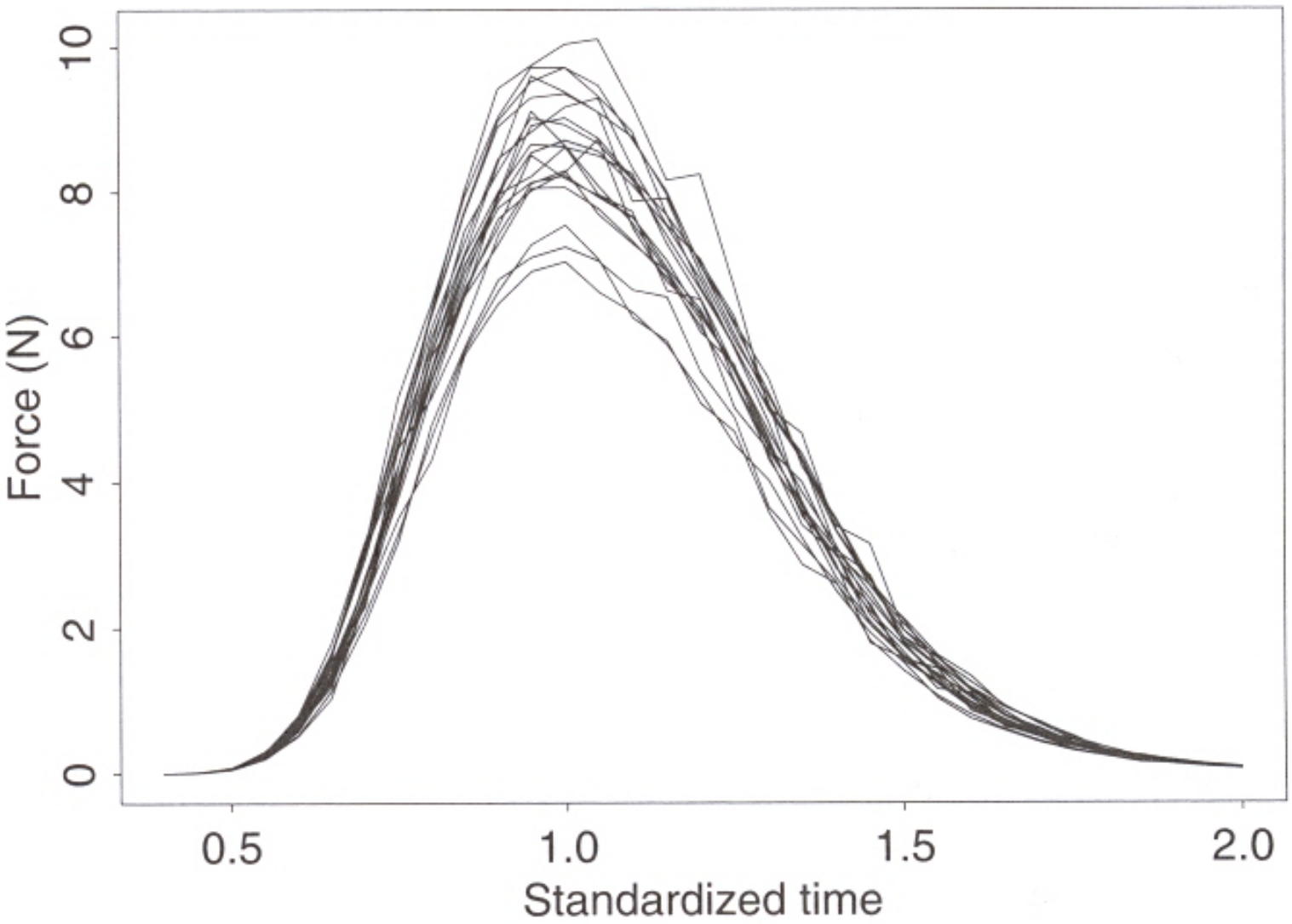
$$\begin{aligned}\text{REGSSE} &= \sum_{i=1}^{N} \int_{\mathcal{T}} [x_i(t+\delta_i) - \hat{\mu}(t)]^2 dt \\ &= \sum_{i=1}^{N} \int_{\mathcal{T}} [x_i^*(t) - \hat{\mu}(t)]^2 dt.\end{aligned}$$

- The optimal $\delta_i = \text{argmin}_\delta \int_{\mathcal{T}} [x_i(t+\delta) - \hat{\mu}(t)]^2 dt$.

Iterations of the least square criterion

- We then iterate this process, by re-computing the mean $\hat{\mu}(t)$ from the registered curves $x_i^*(t)$, and re-computing a new set of shifts δ_i.
- These iterations usually converge in a few cycles.

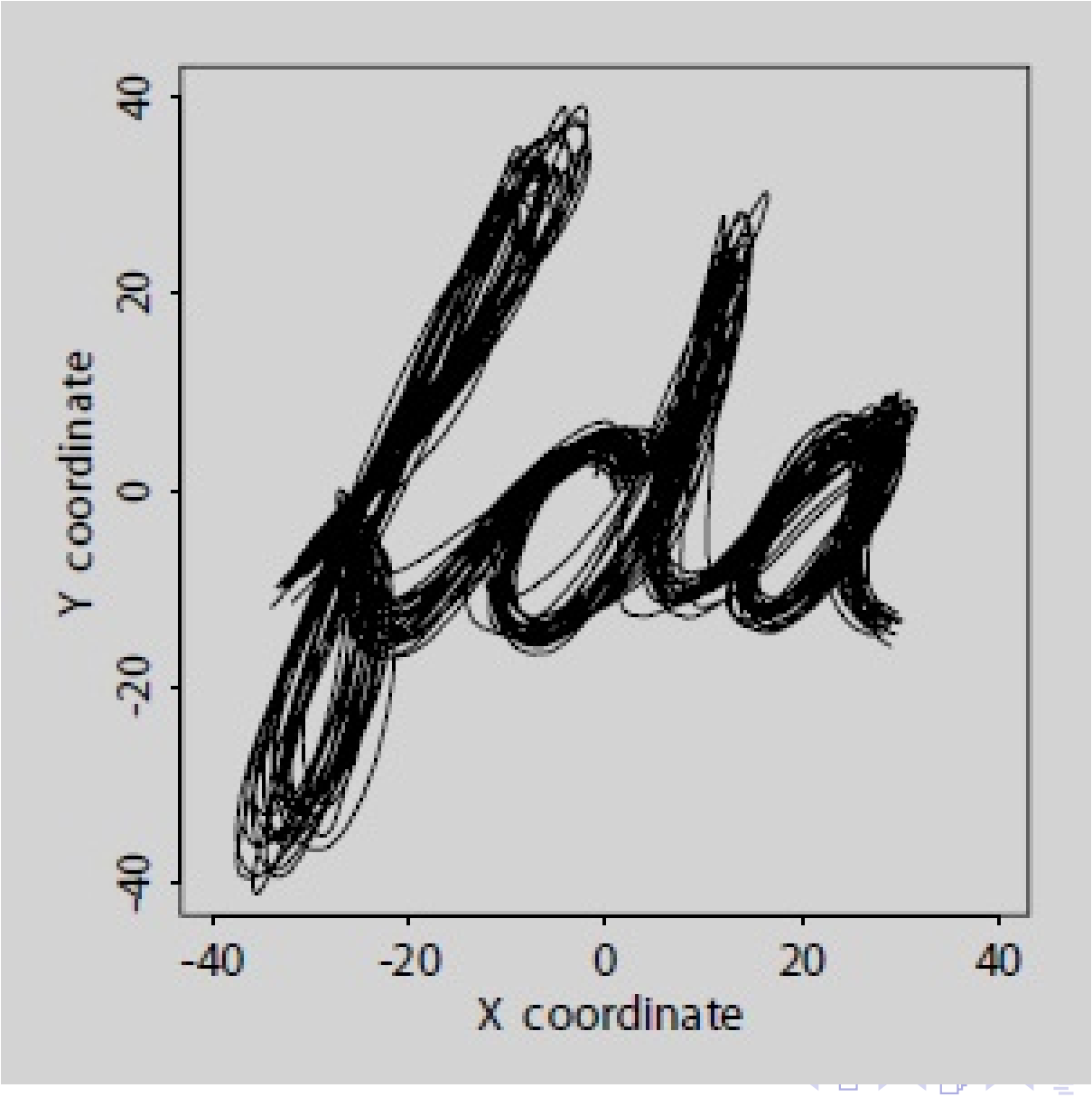
Pinch force data after registration



Feature registration

- A *landmark* or a *feature* of a curve is some characteristic that one can associate with a specific argument value t.
- These are typically maxima, minima, or zero crossings of curves.
- They may be identified with zeros at the level of some derivatives as well as at the level of the curves themselves.
- For example, we may want to align both the summer maximum temperature and the winter minimum at the same time.
- A simple time shift will seldom achieve this. For each curve x_i we identify the argument values $t_{if}, f = 1, \ldots, F$, associated with each of F features.

Handwriting data

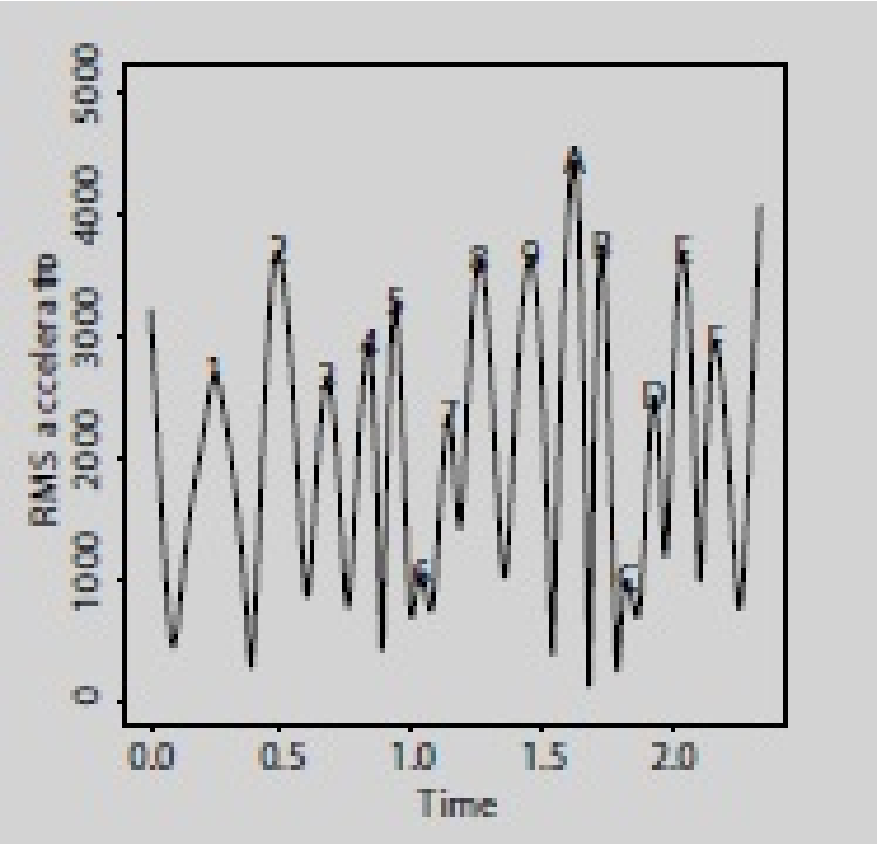


Handwriting data example

- Each sample of handwriting was obtained by recording the pen position 600 times per second.
- There was some preprocessing to make each script begin and end at times 0 and 2.3 seconds, and to compute coordinates at the same 1,401 equally-spaced time values.
- Each curve x_i in this situation is vector-valued, since two spatial coordinates are involved.
- We use ScriptX_i and ScriptY_i to designate the X- and Y-coordinates, respectively.

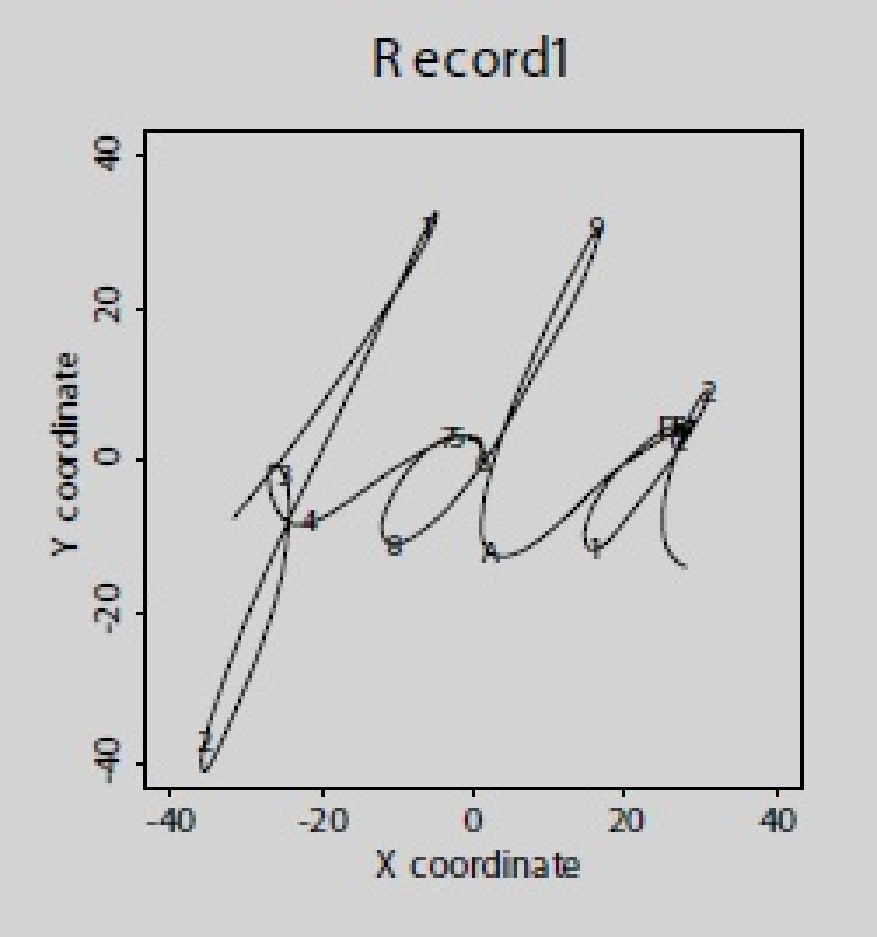
Features in acceleration of handwriting example

The average length of the acceleration vector for the 20 handwriting samples. The characters identify the 15 features used for landmark registration.



Features in Record 1

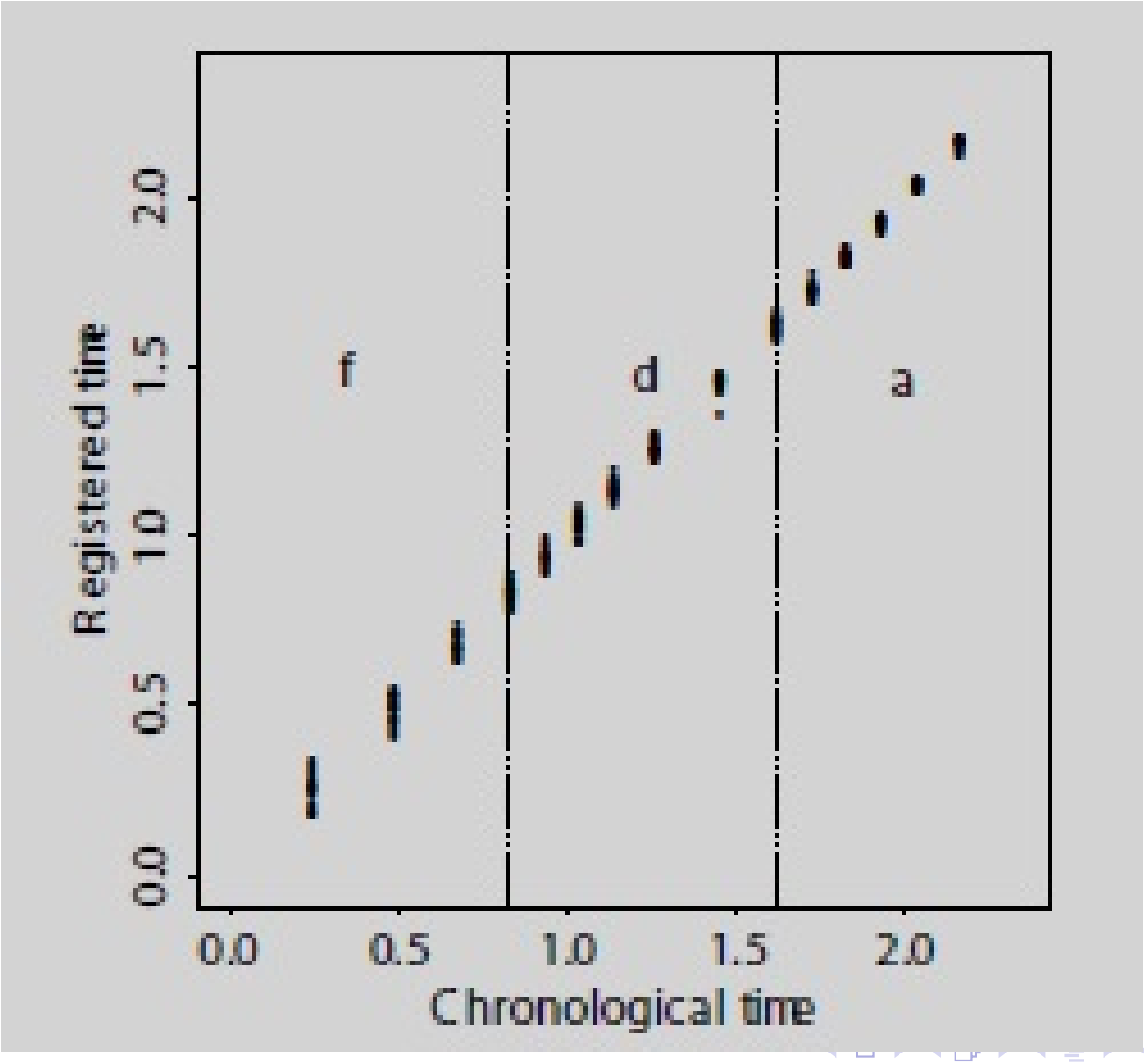
Positions of landmarks in Record 1.
Landmarks "4" and "A" mark the boundaries between letters.



Aligning features with a warping function

- We want to construct a time transformation $h_i(t)$ for each curve such that the registered curves with values
$$x_i^*(t) = x_i[h_i(t)]$$
have identical argument values for any given landmark.
- We refer to $h_i(t)$ as the *time-warping function*. It has the properties
  - $h_i(0) = 0$
  - $h_i(2.3) = 2.3$
  - $h_i(t_{0f}) = t_{if}, f = 1, \ldots, 15$; t_{0f}: reference times
  - h_i is strictly increasing: $s < t$ implies that $h_i(s) < h_i(t)$.

Sample vs mean feature times



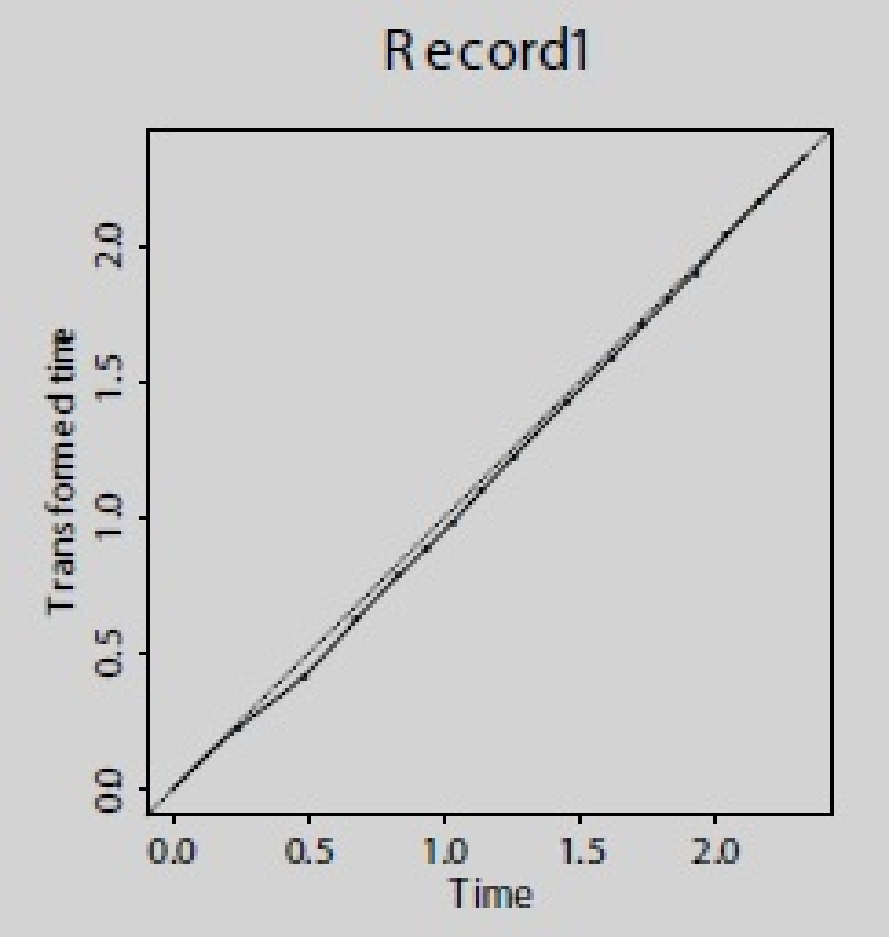
Warping function in handwriting example

In this application,

- besides aligning up the individual feature times t_{if} to t_{0f},
- we used linear interpolation for time values between the points (t_{0f}, t_{if}), as well as the boundary points $(0, 0)$ and $(2.3, 2.3)$, to define the time warping function h_i for each curve.

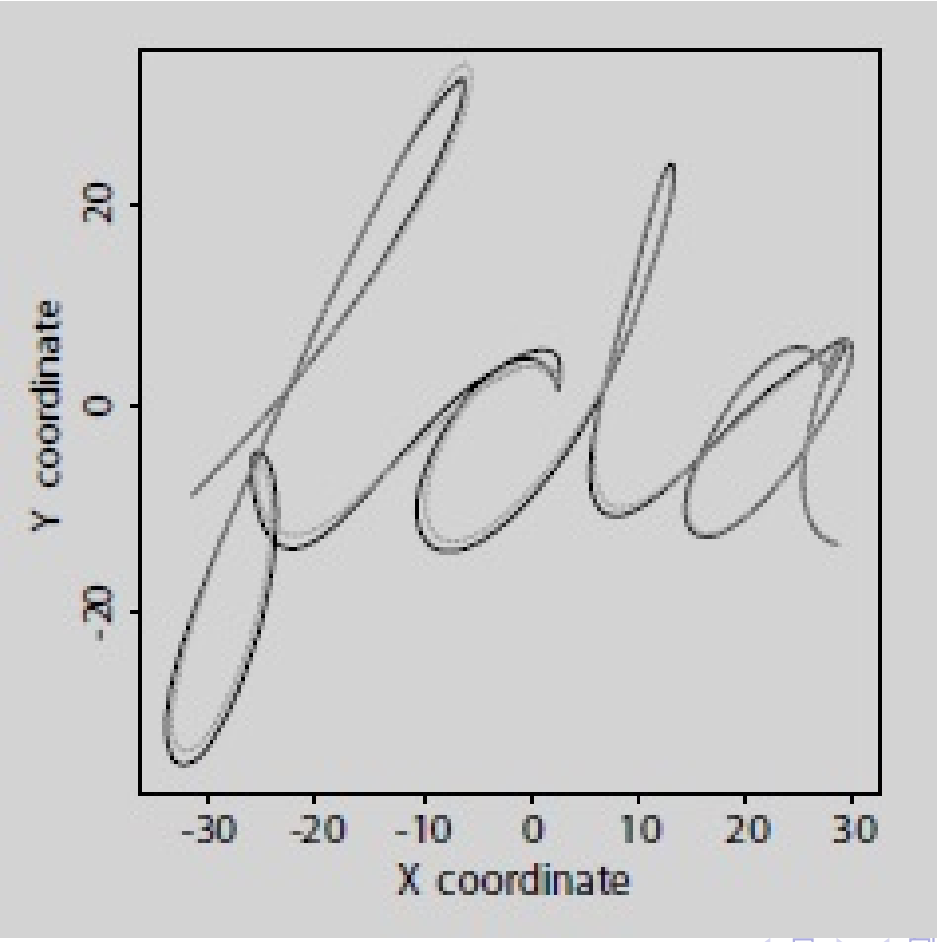
Warping function in Record 1

The time warping function h_1 estimated for the first record that registers its features with respect to the mean curve.



Registered vs unregistered mean curves

Solid line: the mean of the registered "fda" curves (more pronounced loops in "f", better representation?).
Dashed line: the mean of the unregistered curves.



Register x using the warping function

We want to evaluate the registered function $x^*(\mathbf{t}) = x[h(\mathbf{t})]$ at a vector $\mathbf{t}$ of values.

- Compute $h^{-1}(t)$ by smoothing or interpolating the plot of t against $h(t)$. Need *strictly monotone* smoothing.
- Compute $x^*(\mathbf{t})$ after smoothing or interpolating the relationship of $x(t)$ over $h^{-1}(t)$.

A more general warping function

- Time is a growth process, and thus can be expressed by the strictly monotone curves that we used for the children's growth curves.
- That is,

$$h(t) = C_0 + C_1 \int_0^t \exp[W(u)]du.$$

- Constants C_0 and C_1 are fixed by the requirement that $h(t) = t$ at the lower and upper limits of the interval over which we model the data.
- Or, if shift registration is a possibility (i.e. periodic data), the constant term C_0 can be allowed to pick any constant phase shift that is required.

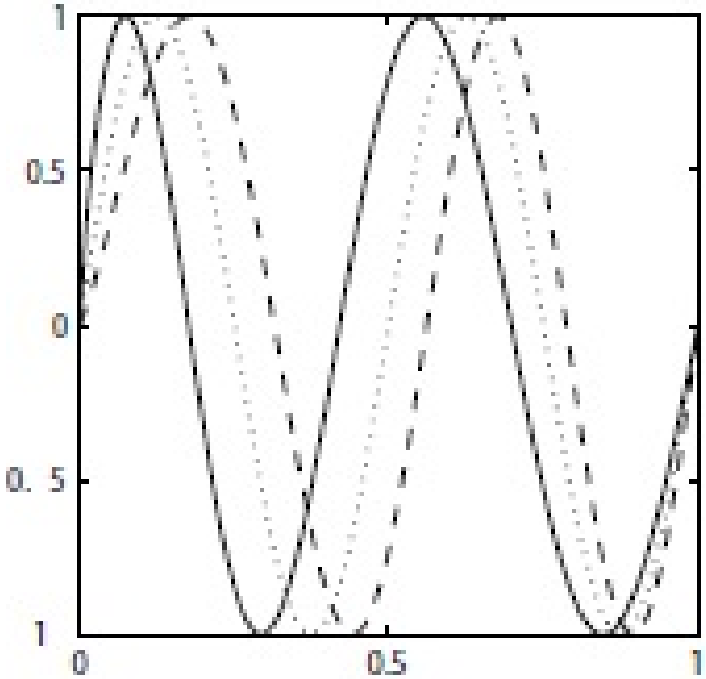
About the general warping function

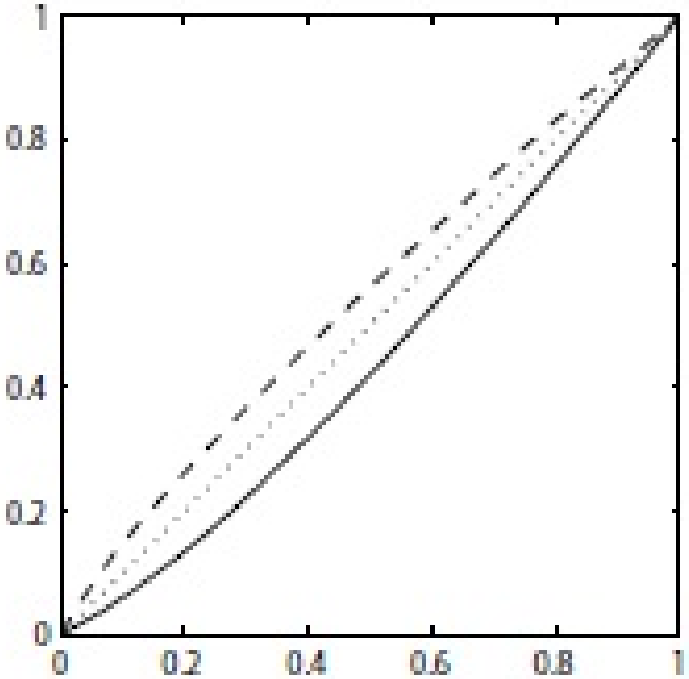
$$h(t) = C_0 + C_1 \int_0^t \exp[W(u)]du.$$

- Physical or clock time grows linearly, and thus corresponds to $W(u) = 0$.
- If $W(u)$ is positive, then $h(t) > t$, warped time is growing faster than clock time, and the observed process is *running late*.
- Negative values of $W(u)$ indicates $h(t) < t$, clock time is being slowed down for a process that is *running ahead* of some target.
- Providing that the warp h is reasonably smooth, the inverse warp h^{-1} is achieved to a close approximation by merely replacing W in the equation by $-W$.

An example with general warping functions

Dotted line: target function $x_0(t) = \sin(4\pi t)$.
Solid lines: an *early* function $x_E(t) = \sin(4\pi t^{0.8})$.
Dashed lines: a *late* function $x_L(t) = \sin(4\pi t^{1.2})$.



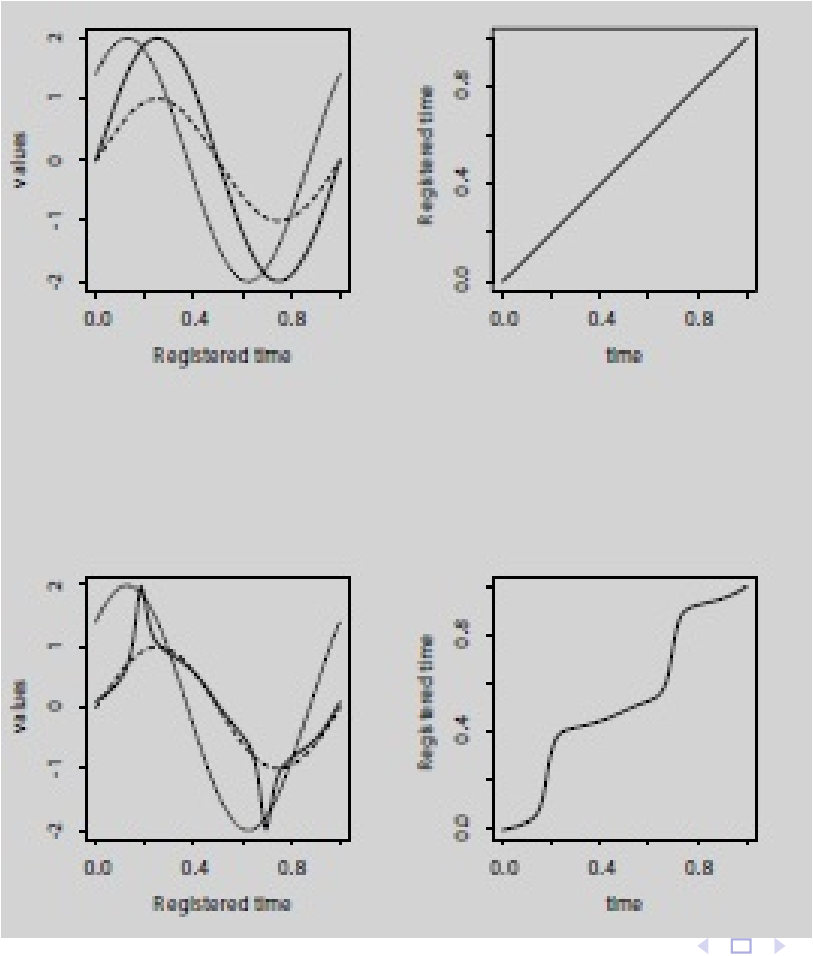


A problem with the LS criterion

- The least squares fitting criterion is intrinsically designed to assess differences in amplitude rather than phase.
- Recall that the mean function is a least squares estimate.
- When two functions differ in terms of amplitude as well as phase, the least squares criterion can use time warping to also minimize amplitude differences by trying to squeeze out of existence regions where amplitudes differ.
- This wasn't a problem when only time shifts were involved since such simple time warps cannot affect amplitude differences.

An example of the LS problem

Upper: PCA-based criterion. Bottom: LS criterion. Dotted: curve to be registered. Dashed: alignment standard. Solid: registered curve.



Using PCA to define a registration criterion

- Suppose two curves x_0 and x_1 differ only in amplitude but not in phase.
- For example, let $x_0(t) = Ax_1(t), A > 0$.
- Then, if we plot the function values $x_0(t)$ and $x_1(t)$ against each other, we will see a straight line.
- Amplitude differences will then be reflected in the slope A of the line, a line at 45° corresponding to no amplitude difference.

Using PCA to define a registration criterion

- Let us consider now evaluating both the target function x_0 and the registered function x^* at a fine mesh of n values of t to obtain the pairs of values $(x_0(t), x[h(t)])$.
- Let the $n \times 2$ matrix $\mathbf{X}$ contain these pairs of values.
- Then the two-by-two cross-product matrix $\mathbf{X}'\mathbf{X}$ would be what we would analyze by principal components.
- A principal components analysis of lines such as these will reveal a second eigenvalue at or near 0.

Using PCA to define a registration criterion

- The following order two matrix is the functional analogue of the cross-product matrix $\mathbf{X}'\mathbf{X}$.

$$\mathbf{T}(h) = \begin{bmatrix} \int \{x_0(t)\}^2 dt & \int x_0(t) x[h(t)] dt \\ \int x_0(t) x[h(t)] dt & \int \{x[h(t)]\}^2 dt \end{bmatrix}.$$

- The summations over points implied by the expression $\mathbf{X}'\mathbf{X}$ have here been replaced by integrals.
- We have expressed the matrix as a function of warping function h to remind ourselves that it does depend on h.

The minimum eigenvalue criterion

- We can now express our fitting criterion for assessing the degree to which two functions are registered as follows:

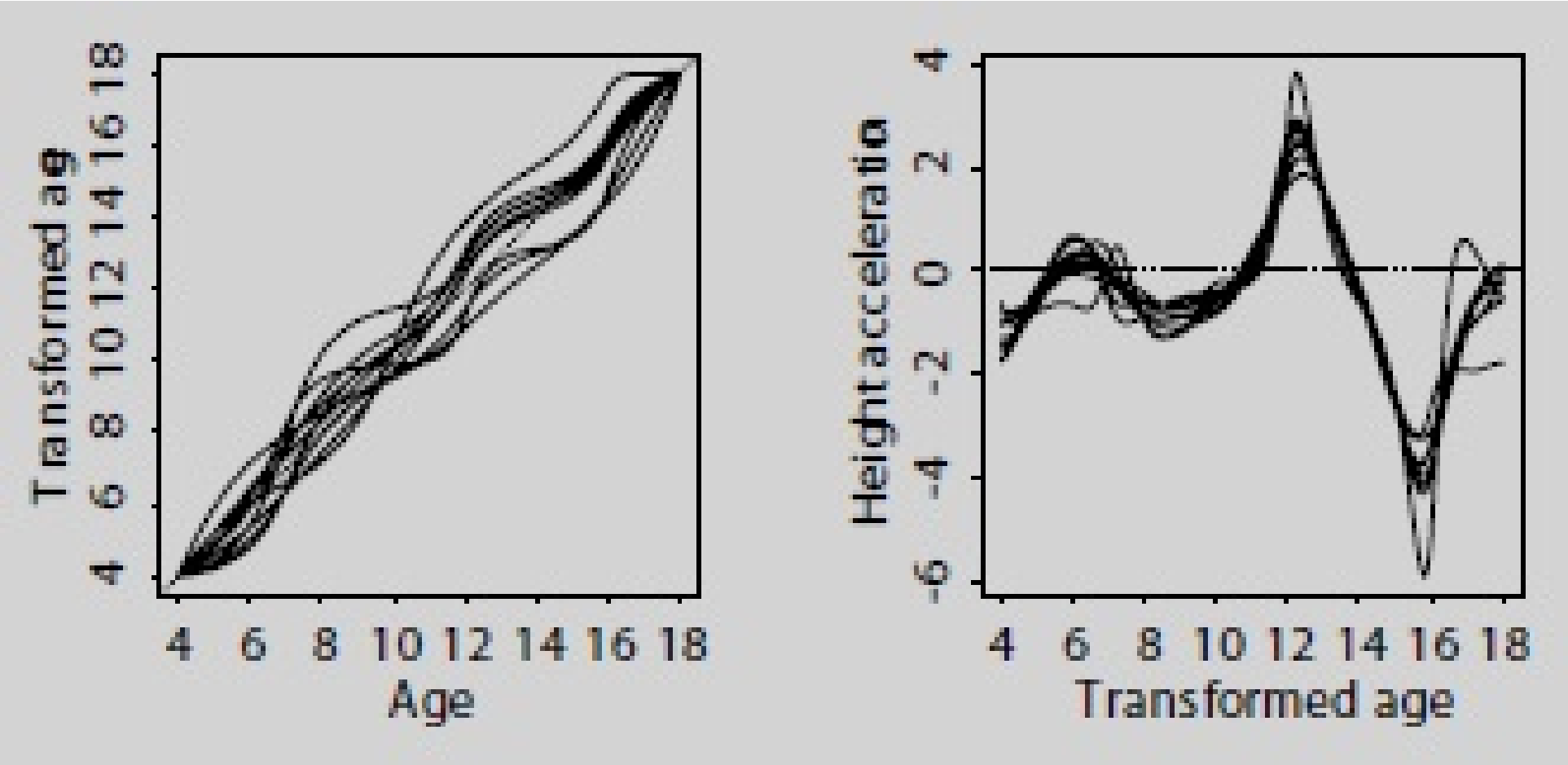
$$\text{MINEIG}(h) = \mu_2[\mathbf{T}(h)],$$

  where the function μ_2 is the size of the second eigenvalue of its argument.
- When $\text{MINEIG}(h) = 0$, we have achieved registration, and h is the warping function that does the job.
- We will want to apply some regularization to impose smoothness on h, so we extend our criterion to

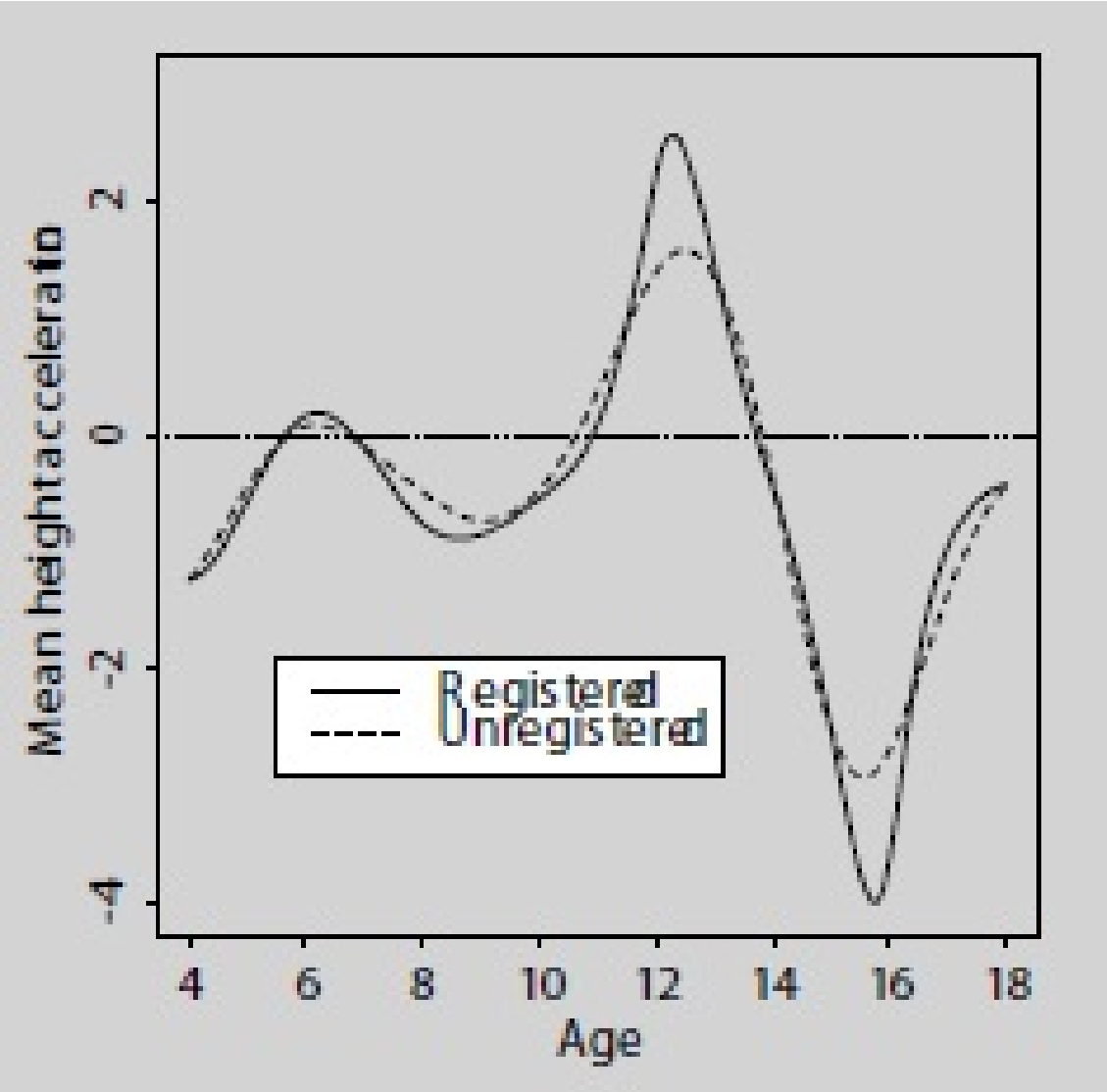
$$\text{MINEIG}_\lambda(h) = \text{MINEIG}(h) + \lambda \int \{W^{(m)}(t)\}^2 dt.$$

Registered height acceleration curves

Left: Estimated time-warping functions.
Right: Registered curves.



Registered vs unregistered cross-sectional means



- Before moving on to more sophisticated procedures, remove amplitude effects that can be accounted for by vertical shifts or scale changes, by centering or possibly rescaling the curves.
- Zero crossing are easier to align. So aligning derivatives might be a better option than aligning the original curves due to their zero crossings.
- It is often wise to first register on any clearly identifiable landmarks before using the continuous registration procedure.